

Midterm

Instructions

This exam is **30 pages** long, and will last **120 minutes**. Before the exam, please read this entire page, then answer the question at the bottom. Do **not** open the test until instructed!

- All outside resources are prohibited, including but not limited to, books, other people, or anything with internet access. Sole exception is one hand-written, double-sided, letter-sized cheat sheet.
- The problems are listed in no particular order of difficulty. Make sure to read all of the problems before getting started.
- There are blank pages at the end of the exam for scratch work. If you would like work on that page to be graded, please clearly indicate so on the problem. **Do not tear them off.**
- Do not do more than what is asked of you. For example, if we do not ask for a proof of correctness for your algorithm, you do not need to provide one. Indecipherable answers will receive no credit.
- Any algorithm covered in lecture may be used as a black box, which means you can include it in your algorithms and talk about its correctness/runtime without proof.
- After the exam, please **do not (publicly) discuss the exam material** until solutions are released. There are some students taking the exam on a later date!

Good luck! You got this.

(a) Please write your name below.

(b) Please write your ChicagoID (e.g. 12345678A) below.

[You can use this page as part of your solution or scratch paper]

PROBLEM 1 (Asymptotics - 15 points). For each pair of functions $f(n), g(n)$, identify if $f(n) = O(g(n))$, $f(n) = \Omega(g(n))$, or $f(n) = \Theta(g(n))$. Pick at most one of the three options for each case. Additionally, pick the most specific option. Explanations may receive partial credit, but they are not required to receive full credit.

	$f(n)$	$g(n)$	$f(n) = O(g(n))$	$f(n) = \Omega(g(n))$	$f(n) = \Theta(g(n))$
a)	$n + n^2$	$n^2 - n$	☐	☐	☐
b)	$\sqrt{n}$	$2^{\log n}$	☐	☐	☐
c)	$\frac{n}{\log n}$	$n^{0.99}$	☐	☐	☐
d)	$7n$	$n \log n$	☐	☐	☐
e)	$n!$	$2^{(n^2)}$	☐	☐	☐

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PROBLEM 2 (Recurrence Relations – 12 points). Write down the solution to the following recurrence relations in the boxes provided. Only the final answer will be graded. For each part, provide the most specific big-O bound possible. You may assume that $T(0) = T(1) = 1$.

(a) $T(n) = 27 \cdot T(n/3) + O(n^3)$

(b) $T(n) = 2 \cdot T(n-1) + 2$

(c) $T(n) = T(n/4) + n/2$

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PROBLEM 3 (Middle elements - 12 points). Suppose that we have an **unsorted** array of integers $A[1 \dots 3n]$ that does not contain duplicates. Our goal is to find an **element** $A[i]$ that is bigger than at least n elements and smaller than at least n elements of A . If A was sorted then $A[i]$ would be in one of the positions $\{n+1, n+2, \dots, 2n\}$.

Q1. Here is a randomized algorithm for this problem.

```

Algorithm1(A[1...3n]):
    j = RANDOM(1, 3n)
    below = 0
    above = 0

    // Count elements over and under A[j]
    for i = 1...3n
        if (A[i] < A[j])
            below += 1
        else if (A[j] < A[i])
            above += 1

    // Check success
    if (above > n) and (below > n)
        return A[j]
    else
        return Algorithm1(A[1...3n])

```

(a) (3 points) What is Algorithm1's **expected** running time? Briefly explain.

(b) (3 points) What is the success probability?

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Q2. Here is another randomized algorithm for this problem. MEDIAN3 takes in three values and returns the middle one.

```
Algorithm2(A[1...3n]):  
    i = RANDOM(1, 3n)  
    j = RANDOM(1, 3n)  
    k = RANDOM(1, 3n)  
    return MEDIAN3(A[i], A[j], A[k])
```

(a) (3 points) What is Algorithm2's **expected** running time? Briefly explain.

(b) (3 points) What is the success probability?

(Hint: recall what you learned from Discrete Math. Each number is drawn independently.)

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PROBLEM 4 (Proofs on DAGs - 17 points). Let $G = (V, E)$ be a DAG with n vertices (i.e. $|V| = n$).

- (a) (8 points) Give a formal proof that $|E| \leq \binom{n}{2}$. This is saying that any DAG with n vertices cannot have more than $\binom{n}{2}$ edges. (Hint: try induction)

- (b) (3 points) Show that the above bound is tight. In another word, show that for every n , there exists a DAG with n vertices and $\binom{n}{2}$ edges.

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- (c) (6 points) Given $G = (V, E)$ is a DAG with n vertices and $\binom{n}{2}$ edges, give a formal proof that there exists an *ordering* of the vertices $f : V \rightarrow \{1, \dots, n\}$, such that for any vertex v , the longest directed path starting at vertex v has exactly $n - f(v)$ edges.

(Hint: think about how DFS produces the topological sort on a DAG)

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PROBLEM 5 (Set convolutions - 18 points). Let A and B be sets of integers, each with n elements, i.e., $|A| = |B| = n$. Define the sum $A + B$ as the set:

$$A + B = \{z \in \mathbb{Z} : z = a + b, a \in A, b \in B\},$$

i.e., the set of integers z which can be written as a sum of an integer in A and one in B . For example, for $n = 3$ when $A = [1, 2, 3]$ and $B = [1, 5, 10]$, then $A + B = [2, 3, 4, 6, 7, 8, 10, 11, 12]$. We are interested in computing $|A + B|$, the number of elements in $A + B$. In the example, that would be $|A + B| = 9$.

You may also use the function $\text{FFT}([a_0, \dots, a_n], \omega)$ which runs FFT given the coefficients $[a_0, \dots, a_n]$ of a degree- $n - 1$ polynomial, and the n -th root of unity ω .

(a) (3 points) Give a small example ($n \leq 3$) of a case where $|A + B| \neq n^2$.

(b) (3 points) Describe a brute-force algorithm that only requires at most $O(n^2 \log n)$ arithmetic operations over integers.

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Suppose that all the elements of A and B lie in $[0, M - 1]$, where $M = 2^k$, is a power of 2.

- (c) (10 points) Write **pseudocode** for an algorithm that accepts sets A and B and computes $|A + B|$ while only performing $O(M \log M)$ arithmetic operations over integers.

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(d) (2 points) Compute the running time of your algorithm.

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PROBLEM 6 (Divide & Conquer - 16 points). Let $A = [a_1, a_2, \dots, a_n]$ and $B = [b_1, b_2, \dots, b_n]$ be two **sorted** arrays.

- (a) (4 points) Write **pseudocode** for an algorithm `median4` that takes as input A and B and outputs the **median** value of their concatenation in the base case when $|A| = |B| = n = 2$. You may assume that all elements in the two lists are **distinct**.

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- (b) (8 points) Write **pseudocode** for a **recursive** divide-and-conquer algorithm that takes as input A and B and outputs the **median** value of their concatenation. No credit will be given to solutions that require $\Omega(n)$ comparisons. You may assume that $n = 2^k$ is a **power of 2** and that all elements in the two lists are **distinct**.

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(c) (2 points) Write the recurrence for the number of comparisons your algorithm requires and solve it.

(d) (2 points) Identify an algorithm we've seen in class (lecture, homework, or discussion) that influenced the design of your algorithm. What component(s) of the algorithm are similar?

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